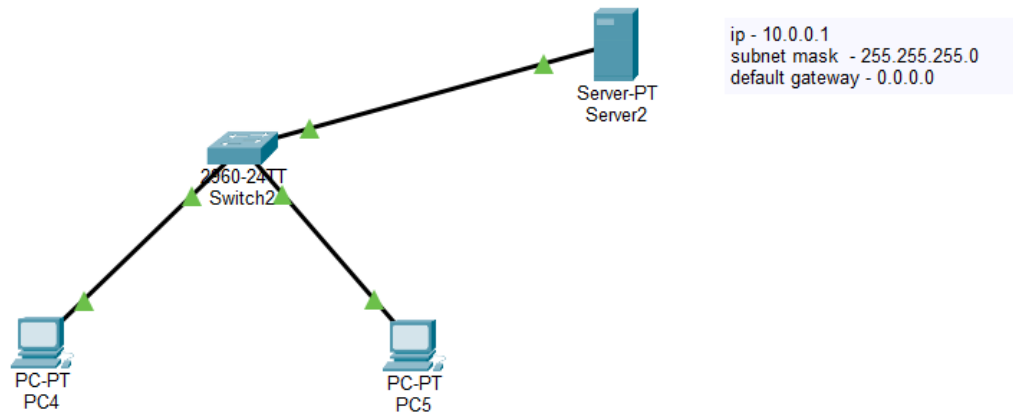


Tasks

1. In Cisco Packet Tracer, set up a simple network with one DHCP server and two client PCs. Configure the DHCP server to assign IP addresses automatically, then verify on each PC that it receives a different IP address from the DHCP pool.



Step 1 : Configure DHCP Service

- Go to **Services** → **DHCP**.
- Turn **DHCP Service ON**.

Field	Value
Pool Name	DHCP_POOL
Default Gateway	10.0.0.1
DNS Server	8.8.8.8
Start IP Address	10.0.0.2
Subnet Mask	255.255.255.0
Maximum Users	50

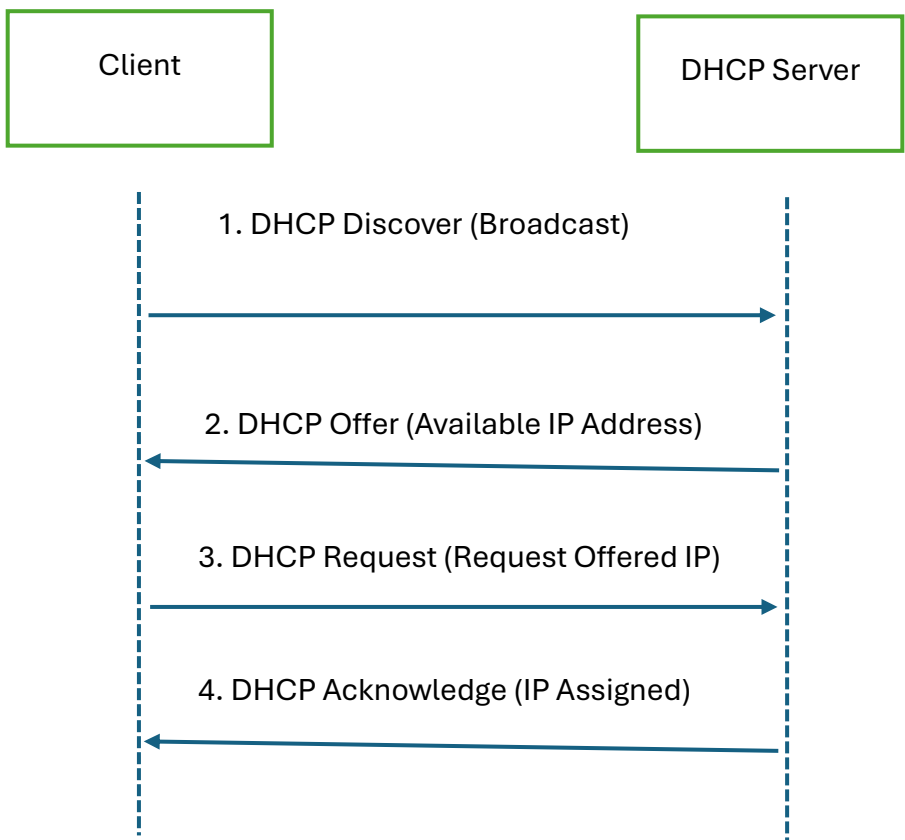
- Click add

Step 2: Configure PC

1. Click **PC**
2. Go to **Desktop** → **IP Configuration**.
3. Select **DHCP**

2. Draw a sequence diagram (hand-drawn or digital) that clearly shows each step of the DHCP process: Discover, Offer, Request, and Acknowledge, labeling the sender and receiver at each stage.

Answer :-



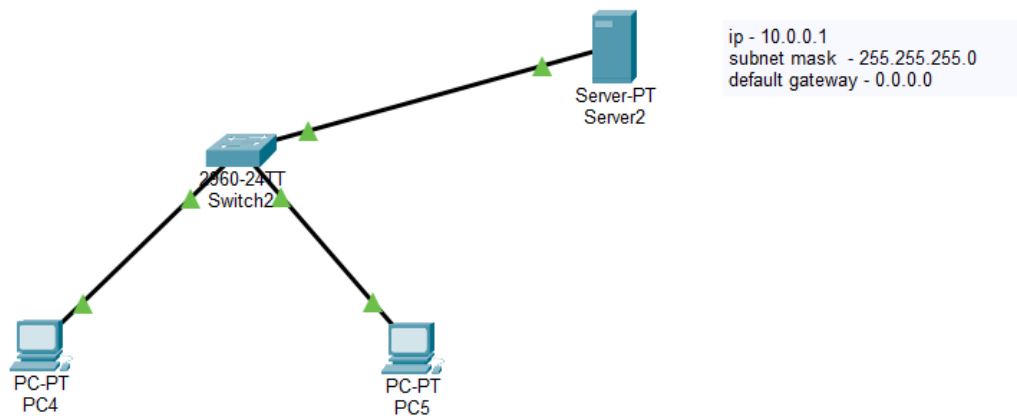
Step	Sender	Receiver	Purpose
DHCP Discover	Client PC	DHCP Server	Client searches for a DHCP server.
DHCP Offer	DHCP Server	Client PC	Server offers an available IP address.
DHCP Request	Client PC	DHCP Server	Client requests the offered IP address.
DHCP Acknowledge (ACK)	DHCP Server	Client PC	Server confirms and assigns the IP address.

The DHCP process follows the **DORA** sequence:

D – Discover → **O** – Offer → **R** – Request → **A** – Acknowledge

After the ACK message, the client receives a valid IP address and can communicate on the network.

3. Change the DHCP pool range on your Cisco Packet Tracer DHCP server so that only one IP address is available. Add two PCs and observe what happens when both try to get an IP address. Take a screenshot of the result and explain in 2-3 lines why only one PC succeeds.



Step 1 : Configure DHCP Service

- Go to **Services** → **DHCP**.
- Turn **DHCP Service ON**.

Field	Value
Pool Name	DHCP_POOL
Default Gateway	10.0.0.1
DNS Server	8.8.8.8
Start IP Address	10.0.0.2
Subnet Mask	255.255.255.0
Maximum Users	1

- Click add

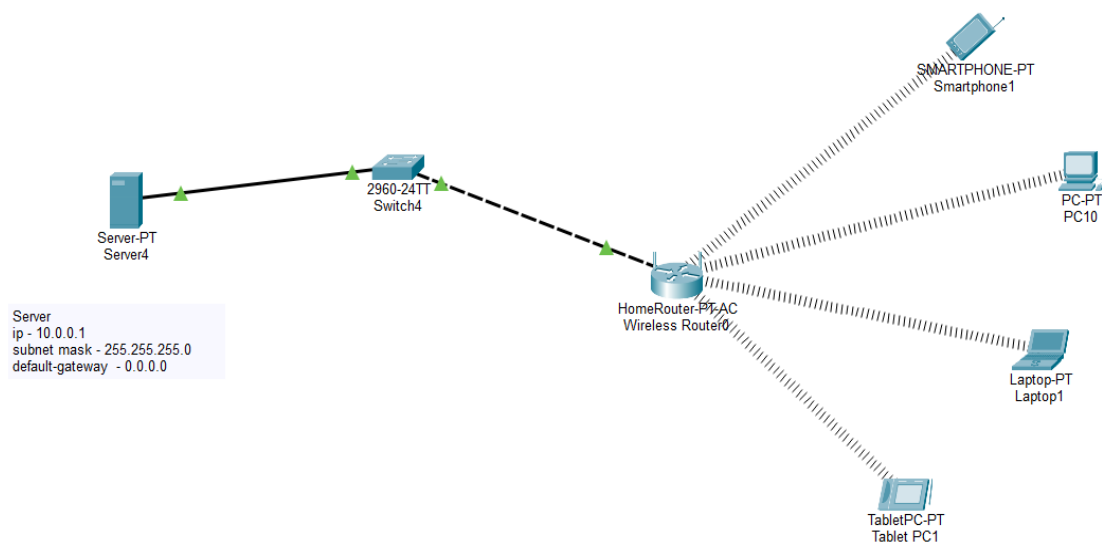
Step 2: Configure PC

4. Click **PC**
5. Go to **Desktop** → **IP Configuration**.
6. Select **DHCP**

The DHCP pool was configured with **Maximum Users = 1**, so only one IP address was available. The first PC received the available IP address, but when the second PC requested an IP address, no addresses were left in the DHCP pool. Therefore, only one PC succeeded in obtaining an IP address.

4. Simulate a Zomato-style restaurant WiFi scenario in Cisco Packet Tracer: configure a DHCP server to provide IP addresses to both a laptop and a smartphone (use a generic PC as smartphone). Show the assigned IPs and explain why DHCP is useful in this situation.

Answer :-



Step 1 : Configure DHCP Service

- Go to **Services** → **DHCP**.
- Turn **DHCP Service ON**.

Field	Value
Pool Name	ZOMATO_WIFI
Default Gateway	10.0.0.1
DNS Server	8.8.8.8
Start IP Address	10.0.0.2
Subnet Mask	255.255.255.0
Maximum Users	50

- Click add

Step 2 : Configure Wireless Router

Wireless → Basic Wireless Settings

Setting	Value
SSID	Zomato_Wifi
SSID Broadcast	Enabled

- Click Save.

Step 3 : Configure Wireless Security

Wireless → Wireless Security

Setting	Value
Security Mode	WPA2-Personal
Encryption	AES
Pre-Shared Key	zomato123

- Save Settings.

Step 4 : Steps to Connect a PC to WiFi

- Click on the PC.
- Open the Physical tab.
- Turn the PC OFF.
- Insert the WMP300N wireless card.
- Turn the PC ON.
- Open Desktop → PC Wireless.
- Select Zomato_Wifi.
- Enter the password (zomato123).
- Click Connect.
- Open Desktop → IP Configuration.
- Select DHCP.
- The PC will receive an IP address automatically.

Step 5 : Get IP

- Desktop → IP Configuration
- Select **DHCP**